

# Buchsbaumness, Macaulayfication and Castelnuovo–Mumford Regularity of Monomial Curves

Biplab Dawn

Indian Institute of Technology Patna

Value Semigroups and Good Semigroups  
University of Catania, Italy  
17 June 2026

Joint with Kumari Saloni (IIT Patna)

arXiv:2506.20473

# Outline

Preliminaries

Projective Monomial Curves and Their Invariants

Our Results

References

## Homogenization

Let  $g(x_1, \dots, x_n) \in \mathcal{K}[x_1, \dots, x_n]$  be a polynomial of total degree  $d$ . Let  $g = \sum_{i=0}^d g_i$  be the expansion of  $g$  as the sum of its homogeneous components, where  $g_i$  has total degree  $i$ . Then the polynomial

$$g^h(x_0, \dots, x_n) = \sum_{i=0}^d g_i(x_1, \dots, x_n) x_0^{d-i}$$

is a homogeneous polynomial of total degree  $d$  in  $\mathcal{K}[x_0, \dots, x_n]$ . We call  $g^h$  the *homogenization* of  $g$ .

## Homogenization of an Ideal

Let  $I$  be an ideal in  $\mathcal{K}[x_1, \dots, x_n]$ . We define the *homogenization* of  $I$  to be the ideal

$$I^h = \langle f^h \mid f \in I \rangle \subseteq \mathcal{K}[x_0, \dots, x_n],$$

where  $f^h$  is the homogenization of  $f$ .

## Graded Rings

A *graded ring* is a ring  $A$  together with a family  $\{A_n\}_{n \geq 0}$  of subgroups of the additive group of  $A$ , such that

$$A = \bigoplus_{n=0}^{\infty} A_n \quad \text{and} \quad A_m A_n \subseteq A_{m+n} \text{ for all } m, n \geq 0.$$

## Graded Modules

Let  $A = \bigoplus_{n \geq 0} A_n$  be a graded ring. A *graded  $A$ -module* is an  $A$ -module  $M$  together with a family  $\{M_n\}_{n \in \mathbb{Z}}$  of subgroups of the additive group of  $M$ , such that

$$M = \bigoplus_{n \in \mathbb{Z}} M_n$$

and

$$A_m M_n \subseteq M_{m+n} \quad \text{for all } m \geq 0, n \in \mathbb{Z}.$$

## Projective closure

Given an affine variety  $W \subseteq \mathcal{K}^n$ , the *projective closure* of  $W$  is the projective variety

$$\overline{W} = V\left(I_a(W)^h\right) \subseteq \mathbb{P}_{\mathcal{K}}^n,$$

where  $I_a(W)^h \subseteq \mathcal{K}[x_0, \dots, x_n]$  is the *homogenization* of the ideal  $I_a(W) \subseteq \mathcal{K}[x_1, \dots, x_n]$ .

## Theorem

Let  $\mathcal{K}$  be an algebraically closed field, and let  $I \subseteq \mathcal{K}[x_1, \dots, x_n]$  be an ideal. Then

$$V(I^h) \subseteq \mathbb{P}_{\mathcal{K}}^n$$

is the *projective closure* of  $V_a(I) \subseteq \mathcal{K}^n$ .

Let  $\mathcal{K}$  be a field, a curve  $\mathcal{C}$  is a one-dimensional closed subscheme of  $\mathbb{P}^n$  which is locally Cohen-Macaulay and equidimensional. Let

$$M(\mathcal{C}) = \bigoplus_{m \in \mathbb{Z}} H^1(\mathbb{P}_{\mathcal{K}}^n, I_{\mathcal{C}}(m))$$

be the Hartshorne-Rao module of  $\mathcal{C}$ . Here  $I_{\mathcal{C}}$  is the ideal sheaf of  $\mathcal{C}$ .

**Theorem**(A. V. Geramita, P. Maroscia, and W. Vogel, 1985)

$M(\mathcal{C}) \cong H_{\mathfrak{M}}^1(R)$  where  $R = \mathcal{K}[x_0, \dots, x_n]/I(\mathcal{C})$ ,  $I(\mathcal{C})$  is the homogeneous ideal of  $\mathcal{C}$ ,  $\mathfrak{M} = (x_0, \dots, x_n)R$  and  $H_{\mathfrak{M}}^1(R)$  is the first local cohomology module with respect to  $\mathfrak{M}$ .

## Definition

We say  $\mathcal{C}$  is strictly  $k$ -Buchsbaum if  $k$  is the least number such that  $\mathfrak{M}^k H_{\mathfrak{M}}^1(R) = 0$  and  $k$ -Buchsbaum if  $\mathfrak{M}^k H_{\mathfrak{M}}^1(R) = 0$ . We say that  $\mathcal{C}$  is arithmetically Cohen-Macaulay iff  $k = 0$ .

## Projective Monomial Curve

Projective monomial curves are defined by their generic zero  $(s^d, s^{d-a_1}t^{a_1}, \dots, s^{d-a_n}t^{a_n}, t^d)$ , where  $a_1 < a_2 < \dots < a_n < d$  are positive integers and  $\gcd(a_1, a_2, \dots, a_n, d) = 1$ .

$M = \{s^d, s^{d-a_1}t^{a_1}, \dots, s^{d-a_n}t^{a_n}, t^d\}$ . If  $R$  is the homogeneous coordinate ring of the monomial curve, then  $R \cong \mathcal{K}[M]$ .

### Note:

- ▶ We can say this curve is smooth if  $a_1 = 1$ ,  $a_n = d - 1$  and it has at most one singular point if either  $a_1 = 1$  or  $a_n = d - 1$ .
- ▶ There are very few class known for non-Cohen Macaulay rings  $R$  whose corresponding monomial curves are non-smooth.

$R = \mathcal{K}[X_1, \dots, X_d]$  is a polynomial ring with usual grading, and

$$M = \bigoplus_{n \in \mathbb{N}} M_n$$

is a finitely generated graded  $R$ -module.

### Theorem (Hilbert, 1890)

Any such module  $M$  has a **finite graded minimal free resolution**:

$$0 \longrightarrow F_d \longrightarrow \cdots \longrightarrow F_1 \longrightarrow F_0 \longrightarrow M \longrightarrow 0,$$

where

$$F_i = \bigoplus_{j \in \mathbb{Z}} R(-j)^{\beta_{ij}}$$

for some integers  $\beta_{ij}$  (called **graded Betti numbers**).



# Castelnuovo-Mumford Regularity

## Definition (Eisenbud and Goto, 1984)

$$\operatorname{reg}(M) = \max\{j - i : \beta_{ij}(M) \neq 0, 0 \leq i \leq d, j \in \mathbb{Z}\}.$$

- ▶ Let  $R = R_0[x_1, \dots, x_d]$  be a Noetherian **standard**  $\mathbb{N}$ -graded ring.
- ▶ Let  $I$  be a **homogeneous ideal** of  $R$ , and  $M = \bigoplus_{n \in \mathbb{N}} M_n$  is a graded  $R$ -module.
- ▶ The 0th **local cohomology** of  $M$  with respect to  $I$  is defined to be

$$H_I^0(M) := \{x \in M : I^n x = 0 \text{ for some } n \geq 1\}.$$

- ▶ It gives a left exact covariant functor  $H_I^0(-) : \operatorname{Mod}(R) \longrightarrow \operatorname{Mod}(R)$ .
- ▶ The local cohomology functors  $\{H_I^n(-) : n \in \mathbb{Z}\}$  are defined to be the right derived functors of  $H_I^0(-)$ .

## Regularity in terms of local cohomology

Let  $R = R_0[x_1, \dots, x_d]$  be a Noetherian **standard**  $\mathbb{N}$ -graded ring, and  $M = \bigoplus_{n \in \mathbb{N}} M_n$  is a finitely generated graded  $R$ -module.

**Definition (Castelnuovo, 1893; Mumford, 1966)**

- ▶ Set  $R_+ := \bigoplus_{n \geq 1} R_n = (x_1, \dots, x_d)$ .
- ▶ Set  $\text{end}(M) := \sup\{n : M_n \neq 0\}$ .
- ▶ The **Castelnuovo–Mumford regularity** of  $M$  is defined to be

$$\text{reg}(M) := \max \left\{ \text{end}(H_{R_+}^i(M)) + i \mid 0 \leq i \leq \dim(M) \right\}.$$

# Hilbert Polynomials

- ▶ Consider  $R = \mathcal{K}[X_1, \dots, X_d]$ ,  $M = \bigoplus_{n \in \mathbb{N}} M_n$  is a finitely generated graded  $R$ -module.
- ▶ **Hilbert Function.**  $H_M(n) := \dim_{\mathcal{K}}(M_n)$  for every  $n \in \mathbb{N}$ .

## Theorem (Hilbert)

There exists a polynomial  $P_M(X) \in \mathbb{Q}[X]$  and an integer  $n_0 \in \mathbb{N}$  such that  $H_M(n) = P_M(n)$  for all  $n > n_0$ .

The polynomial  $P_M(X)$  is called the **Hilbert polynomial** of  $M$ .

## Theorem (Grothendieck–Serre, 1955)

$$H_M(n) - P_M(n) = \sum_{i=0}^{\dim(M)} (-1)^i \dim_{\mathcal{K}} \left( H_{R_+}^i(M)_n \right) \quad \text{for all } n \in \mathbb{Z}.$$

- ▶ Clearly,  $H_M(n) = P_M(n)$  for all  $n > \text{reg}(M)$ .

## Bounds of Regularity

Let  $R \cong \mathcal{K}[M]$  be the coordinate ring of a projective monomial curve where  $M = \{s^d, s^{d-a_1}t^{a_1}, \dots, s^{d-a_n}t^{a_n}, t^d\}$

### Eisenbud-Goto Conjecture (1984)

$$\operatorname{reg}(R) \leq d - n.$$

### Theorem (L'vovsky, 1996)

$\operatorname{reg}(R) \leq$  sum of the largest and second largest integer gaps.

### Theorem (Hellus, Hoa and Stückerad, 2010)

For smooth curves  $\operatorname{reg}(R)$  is bounded even by a fraction of the largest integer gap.

## Reduction Number

Let  $R$  is a graded ring and  $J$  is a graded ideal of  $R$  then

$$r_J(R) = \inf\{n \in \mathbb{Z} \mid R_{n+1} = JR_{n+1}\}$$

This number is known as **reduction number** of  $R$  with respect to  $J$ .

**Note:** In the projective monomial curve  $r_Q(R) \leq \text{reg}(R)$ , where  $Q = (s^d, t^d)$ .

**Question:**  $r_Q(R) = \text{reg}(R)$ ?

Unfortunately, this equality does not hold in general, even in smooth monomial curves.

## Macaulyfication

Ring  $\tilde{R}$  is called the *Macaulyfication* of  $R$  if  $R \subseteq \tilde{R}$ ,  $\tilde{R}$  is Cohen-Macaulay and  $\tilde{R}/R$  has finite length.

**Note:**  $R$  has a unique finite Macaulyfication  $\tilde{R}$  for any monomial curve

Let  $n \geq 1$ ,  $r \geq 5$  and consider the set of monomials

$$\mathcal{M} = \{s^{a_n}, s^{a_n-b_1}t^{b_1}, s^{a_n-b_2}t^{b_2}, s^{a_n-a_0}t^{a_0}, s^{a_n-a_1}t^{a_1}, \dots, t^{a_n}\} \quad (1)$$

where  $b_1 = r$ ,  $b_2 = 2r - 2$  and  $a_i = 2r + 2i - 1$  for all  $0 \leq i \leq n$ . Let  $\mathcal{R} \cong \mathcal{K}[\mathcal{M}]$

**Theorem**( - , Saloni, 2025)

Suppose  $5 \leq r \leq 2n + 5$ . Then

- (i)  $\mathcal{R}$  is Cohen-Macaulay if  $r$  is odd and  $2n \geq r + 1$ .
- (ii)  $\mathcal{R}$  is non-Cohen-Macaulay Buchsbaum if  $r$  is odd and  $2n < r + 1$ .
- (iii)  $\mathcal{R}$  is non-Cohen-Macaulay Buchsbaum if  $r$  is even.

Let  $R$  be a projective monomial curve and let  $\mathfrak{M}$  denote its unique homogeneous maximal ideal. Now, let  $a$  be a nonzero element of  $R$  and let the primary decomposition of  $(a)$  be given by  $(a) = \bigcap Q_i(P_i)$ , where each  $Q_i(P_i)$  is  $P_i$ -primary. We Define:

$$C(a) = \bigcap_{P_i \neq \mathfrak{m}} Q_i(P_i).$$

Let  $Q(R)$  be the total quotient ring of  $R$ . Define:

$$\tilde{R} = \left\{ \frac{x}{a}; x \in C(a) \right\} = a^{-1}C(a)$$

It is easy to see that  $R \subseteq \tilde{R} \subseteq Q(R)$

**Theorem**( - , Saloni, 2025)

Suppose that  $R$  is a  $k$ -Buchsbaum domain and  $\tilde{R}$  be as above. Then  $H_{\mathfrak{M}}^1(R) \cong \tilde{R}/R$ . Moreover,  $\tilde{R}$  is the Macaulayfication of  $R$ .

### Remark( - , Saloni, 2025)

Let  $\mathcal{R} = \mathcal{K}[\mathcal{M}]$  and suppose that  $s^{3a_n-3b_1}t^{3b_1} \notin (s^{a_n})$ . Then, for a positive integer  $k'$ ,

$$(s^{k'a_n}) = (s^{k'a_n}, s^{(k'-1)a_n}s^{3a_n-3b_1}t^{3b_1}) \cap (s^{k'a_n}, t^{ma_n})$$

is a primary decomposition of the ideal  $(s^{k'a_n})$  where  $m \in \mathbb{Z}^+$ .

Consequently,  $\widetilde{\mathcal{R}} = \mathcal{K}[\widetilde{\mathcal{M}}]$  is the Macaulayfication of  $\mathcal{R}$  where  $\widetilde{\mathcal{M}} = \mathcal{M} \cup \{s^{2a_n-3b_1}t^{3b_1}\}$ .



Let  $R \cong \mathcal{K}[M]$  be the coordinate ring of a projective monomial curve. Let  $\tilde{R}$  be the Macaulayfication of  $R$ , given by  $\tilde{R} = \mathcal{K}[\tilde{M}]$ , where  $\tilde{M} = M \cup N$  suppose  $N$  consists of monomials of the same degree, say  $ld$ , for some positive integer  $l$ .

We have that  $R$  has standard grading with the monomials in  $M$  having degree one and  $\deg s^\alpha t^\beta = (\alpha + \beta)/d$  for all  $s^\alpha t^\beta \in R$

Define  $R' = \mathcal{K}[N]$ .  $R'$  has standard grading with the monomials in  $N$  having degree one and  $\deg s^\alpha t^\beta = (\alpha + \beta)/ld$  for all  $s^\alpha t^\beta \in R'$ .

We write  $R = \bigoplus_{i \geq 0} R_i$  and  $R' = \bigoplus_{i \geq 0} R'_i$ .

Let  $r_Q(\tilde{R})$  denote the least integer  $m$  such that  $\tilde{R}_{m+1} = (Q \cdot \tilde{R})_{m+1}$  where  $Q = (s^d, t^d)$ .

## Theorem( - , Saloni, 2025)

Suppose  $R'_2 \subseteq R_{2l}$  or  $R'_2 \subseteq R_l R'_1$ . Then  $R$  is strictly  $k$ -Buchsbaum if and only if  $a(\tilde{R}/R) := \max\{n \in \mathbb{Z} \mid (\tilde{R}/R)_n \neq 0\} = k + l - 1$ .

## Corollary

- ▶ Suppose  $R$  is strictly  $k$ -Buchsbaum and  $R'_2 \subseteq R_{2l}$  or  $R'_2 \subseteq R_l R'_1$ . Then  $\text{reg}(R) = \max\{k + l, r_Q(\tilde{R})\}$ .
- ▶ Suppose  $R$  is a smooth monomial curve. Then  $R$  is strictly  $k$ -Buchsbaum if and only if  $\text{reg}(R) = k + 1$ .
- ▶ Let  $\mathcal{R} = \mathcal{K}[\mathcal{M}]$ . Then  $\text{reg}(\mathcal{R}) = \max\{k + 2, r_Q(\tilde{\mathcal{R}})\}$ .

## Theorem( - , Saloni, 2025)

The ring  $R$  is strictly  $k$ -Buchsbaum if and only if  $kG_R + G_{\tilde{R}} = (k + l)G_R$ , where

$$G_R = \{a \in \mathbb{Z}_{\geq 0} \mid s^{d-a}t^a \in M\}, \quad G_{\tilde{R}} = lG_R \cup \{a \in \mathbb{Z}_{\geq 0} \mid s^{ld-a}t^a \in N\}$$

and  $lG_R$  denotes the set of all sums of  $l$  elements from  $G_R$ .

## Corollary

Suppose  $n \in \mathbb{N}$  and  $r = 2n + 8$  in (1). Then  $\mathcal{R}$  is strictly 2-Buchsbaum.

## Theorem( - , Saloni, 2025)

Let  $\mathcal{R} = \mathcal{K}[\mathcal{M}]$ . Then  $\text{reg}(\mathcal{R}) = r_Q(\mathcal{R})$ , where  $Q = (s^{a_n}, t^{a_n})$ .

Let  $\mathcal{R}$  be same as defined in (1) with  $n = 1$ , i.e.,  $\mathcal{R} = k[\mathcal{M}]$  is in  $\mathbb{P}^4$ . So,

$$\mathcal{M} = \{ s^{2r+1}, s^{r+1}t^r, s^3t^{2r-2}, s^2t^{2r-1}, t^{2r+1} \}$$

### Theorem( - , Saloni, 2025)

Let  $\mathcal{R} = k[\mathcal{M}]$  be in  $\mathbb{P}^4$  and  $k \geq 1$ . Then  $\mathcal{R}$  is strictly  $k$ -Buchsbaum if and only if  $r \in \{ 2 + 3k, 3 + 3k, 4 + 3k \}$ .

### Theorem( - , Saloni, 2025)

If  $\mathcal{R}$  is strictly  $k$ -Buchsbaum in  $\mathbb{P}^4$ , then  $\text{reg}(\mathcal{R}) = k + 2$ .

### Question

If  $\mathcal{R}$  is a strictly  $k$ -Buchsbaum ring for  $k \geq 1$ , is it true that  $\text{reg}(\mathcal{R}) = k + 2$ ?

# References:

-  H. Bresinsky, *Monomial Buchsbaum ideals in  $\mathbb{P}^r$* , Manuscripta Math. 47 (1984), 105-132.
-  H. Bresinsky, P. Schenzel and W. Vogel, *On liaison, arithmetical Buchsbaum curves and monomial curves in  $\mathbb{P}^3$* , J. Algebra 86 (1984), 283-301.
-  D. Cox, J. Little, D. O'Shea and M. Sweedler, *Ideals, varieties and algorithms*, Vol. 3, Springer, New York, 1997.
-  D. Eisenbud, *Commutative algebra: with a view toward algebraic geometry*, Vol. 150, Springer, 2013.
-  D. Eisenbud and S. Goto, *Linear free resolutions and minimal multiplicity*, J. Algebra 88 (1984), 89-133.
-  M. Fiorentini and L. T. Hoa, . *On monomial  $k$ -Buchsbaum curves in  $\mathbb{P}^r$* . Annali dell'Università di Ferrara, 36 (1990), 159-174.

-  A. V. Geramita, P. Maroscia, and W. Vogel, *A note on arithmetically Buchsbaum curves in  $\mathbb{P}^3$* , Matematiche (Catania) 40 (1985), 21-28.
-  S. Goto, *On the Cohen-Macaulayfication of certain Buchsbaum rings*, Nagoya Math. J. 80 (1980), 107-116.
-  S. Goto, *Buchsbaum rings with multiplicity 2*, J. Algebra 74 (1982), 494-508.
-  L. Gruson, R. Lazarsfeld and C. Peskine, *On a theorem of Castelnuovo and the equations defining space curves*, Invent. Math. 72 (1983), 491-506.
-  W. Gröbner, *Über Veronesesche varietäten und deren projektionen*, Archiv der Mathematik, 16 (1965), 257-264.
-  M. Hellus, L. Hoa and J. Stückrad, *Castelnuovo-Mumford regularity and the reduction number of some monomial curves*, Proc. Amer. Math. Soc. 138 (2010), 27-35.

-  L. T. Hoa, *Classification of the Triple Projections of Veronese Varieties*, Math. Nachr. 128 (1986), 185-197.
-  L. T. Hoa, *On monomial  $k$ -Buchsbaum curves in  $\mathbb{P}^3$* . Manuscripta Math. 73 (1991), 423-436 .
-  T. T. G. Lam and N. V. Trung, *Buchsbaumness and Castelnuovo-Mumford regularity of non-smooth monomial curves*, J. Algebra 590 (2022), 313-337.
-  S. L'vovsky, *On inflection points, monomial curves and hypersurfaces containing projective curves*, Math. Ann. 306 (1996), 719-735.
-  F.S. Macaulay, *Algebraic Theory of Modular Systems*, Cambridge Tracts, vol. 19, 1916.
-  M. Nitsche, *Castelnuovo-Mumford regularity and reduction number of smooth monomial curves in  $\mathbb{P}^5$* , Proc. Amer. Math. Soc. 140 (2012), 2937-2944.

-  M. Nitsche, *A combinatorial proof of the Eisenbud–Goto conjecture for monomial curves and some simplicial semigroup rings*, J. Algebra 397 (2014), 47-67.
-  P. Schenzel, *On Veronesean embedding and projections of Veronesean varieties*, Arch. Math. 30 (1978), 391-397.
-  J. Stückrad and W. Vogel, *Buchsbaum rings and applications: an interaction between algebra, geometry and topology*, Vol. 21, Springer, Berlin, 1986.
-  N. V. Trung, *Toward a theory of generalized Cohen-Macaulay modules*, Nagoya Math. J. 102 (1986), 1-49.
-  N. V. Trung, *Projections of one-dimensional Veronese varieties*, Math. Nachr. 118 (1984), 47-67.
-  N. V. Trung, *Reduction exponent and degree bound for the defining equations of graded rings*, Proc. Amer. Math. Soc. 101 (1987), 229-236.





N. V. Trung, *Classification of the double projections of Veronese Varieties*, J. Math. Kyoto Univ. 22 (1983), 567–581.



N. V. Trung and L. T. Hoa, *Affine semigroup and Cohen-Macaulay rings generated by monomials*, Trans. Amer. Math. Soc. 298 (1986), 145-167 .

# Thank You